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SPECIAL REPORT



Transmission dynamics of dengue and chikungunya in a changing climate: do we understand the eco-evolutionary response?

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ABSTRACT

Introduction: We are witnessing an alarming increase in the burden and range of mosquito-borne arboviral diseases. The transmission dynamics of arboviral diseases is highly sensitive to climate and weather and is further affected by non-climatic factors such as human mobility, urbanization, and disease control. As evidence also suggests, climate-driven changes in species interactions may trigger evolutionary responses in both vectors and pathogens with important consequences for disease transmission patterns.

Areas covered: Focusing on dengue and chikungunya, we review the current knowledge and challenges in our understanding of disease risk in a rapidly changing climate. We identify the most critical research gaps that limit the predictive skill of arbovirus risk models and the development of early warning systems, and conclude by highlighting the potentially important research directions to stimulate progress in this field.

Expert opinion: Future studies that aim to predict the risk of arboviral diseases need to consider the interactions between climate modes at different timescales, the effects of the many non-climatic drivers, as well as the potential for climate-driven adaptation and evolution in vectors and pathogens. An important outcome of such studies would be an enhanced ability to promulgate early warning information, initiate adequate response, and enhance preparedness capacity.

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1. Introduction

In the past few decades, we have seen an alarming upsurge of mosquito-borne arboviral diseases in concert with the global spread of their mosquito vectors [1]. The most prominent among those are dengue and chikungunya because of their ability to cause large and periodic epidemics in endemic settings [2–6], particularly affecting densely populated tropical urban areas. Both dengue and chikungunya viruses are transmitted primarily by *Aedes aegypti* and *Aedes albopictus*. Both vectors are highly competent and widely distributed in tropical and subtropical areas [7]. More recently, *Ae. albopictus* has gained a strong foothold in temperate areas by adapting to cooler climates and has led to outbreaks in naïve populations where travel-related virus introduction risk has been on the rise, driven by increasing human mobility and connectivity [8].

Climate change is the defining issue of our time. Climate and weather strongly affect the transmission dynamics of arboviruses [9]. The development rates of viruses and vectors are sensitive to even small changes in temperature and precipitation. Major shifts in the distribution of arboviral diseases are anticipated in response to globally rising temperatures and more variable weather, as projected for the coming decades [10]. Moreover, non-climatic drivers, such as urbanization and other socio-economic factors, and the extent and intensity of prevention and control activities, are likely to have complex effects on the

environmental suitability for virus transmission in an ever-changing climate [8,11–14]. Ultimately, the risk of arboviral disease in a population is determined by the level of preparedness to respond to this dynamic environmental suitability. Preparedness necessitates progress in our nascent understanding of the complex links between short-term environmental variability and long-term anthropogenic climate change and disease risk. This would allow us to predict shifts in disease risk and strategically target surveillance and response efforts to reduce arboviral disease burden, both locally and globally.

Here, we focus on dengue and chikungunya and review the current knowledge base and the challenges in our understanding of arboviral disease risk in a rapidly changing climate and environment. We identify the most critical research gaps that limit the predictive skill of arbovirus risk models and the development of early warning systems and conclude by highlighting several potentially important research directions. By doing so, we aim to stimulate future research and progress in this field.

2. Effects of climate change and variability on arboviral diseases

2.1. Climate varies at multiple timescales

Dengue, chikungunya and several other mosquito-borne diseases are sensitive to climate through its impact on both the

Article highlights

- Mosquito-borne arboviral diseases are on the rise in concert with the global spread of their mosquito vectors.
- Complex interactions between climate modes occur at multiple time-scales, affecting the environmental suitability for proliferation of vectors and transmission of arboviruses.
- Time-lagged and nonlinear relationships exist between local climatic conditions (e.g. temperature, rainfall) and arboviral disease risk.
- A comprehensive analysis should also consider the effects of non-climatic factors to determine the changes in exposure and vulnerability to arboviral diseases in any given population.
- Climate-driven changes in species interactions may trigger evolutionary responses in both vectors and pathogens with important consequences for disease transmission patterns.
- Future studies considering these different aspects are urgently needed to predict disease risk across space and time to provide early warning information and enhance preparedness and response capacities on the ground.

vectors (e.g., extrinsic incubation period, habitat suitability) and the pathogens (e.g., virus replication rates) [14–18]. To better comprehend the role of climate variability and change on disease transmission, it is important to understand how climate varies at multiple timescales.

The climate system is defined by the interactions of its five constituent sub-systems: the atmosphere, hydrosphere, cryosphere, lithosphere and biosphere [19]. These interactions tend to be complex and occur at multiple timescales, with physical processes ranging from minutes to days (weather), to several weeks (intra-seasonal), to several months (seasonal), to a few years (inter-annual), to multiple decades (inter-decadal), and centuries (climate change) [20–23].

Interactions can happen between climate modes at one particular timescale. For example, dipolar (opposite sign) sea-surface temperature configurations between the Pacific and the Atlantic oceans modify the behavior of weather patterns that can alter temperatures and rainfall amounts and frequencies in the Americas, enhancing or inhibiting suitable environmental conditions for vectors to transmit dengue or chikungunya.

More generally, interactions between climate modes are, however, not confined to one timescale: they can interfere with other phenomena acting at different timescales [24,25], amplifying or diminishing the original climate signal. For example, the most important intra-seasonal climate mode of variability in the tropics, the Madden-Julian oscillation [26], is known to interact [27,28] with El Niño-Southern Oscillation (ENSO) [29], the main mode of variability at seasonal-to-interannual timescales, modifying the behavior of weather patterns around the world (not only the tropics) contributing or not to vector proliferation. Nonetheless, the particular details of how these cross-timescale interactions impact transmission dynamics require further studies.

2.2. Links between weather and climate and disease transmission and risk

Evidence from laboratory and field settings provides insight into the processes underlying the climate sensitivity of dengue and chikungunya virus transmission by *Aedes* vectors. Vectorial

capacity incorporates important mechanisms for virus transmission in vectors and is closely linked to the estimates of the basic reproduction number of a disease (R_0) [30]—that is, the number of secondary cases originating from one infectious person in a totally naïve population. Temperature – particularly its influence on vector longevity, which determines the ability of vectors to survive long enough to replicate and transmit the virus – is one of the most important and extensively studied climatic drivers [9,10,31]. Another temperature dependent mechanism regulating virus replication in vectors is described by vector competence [8]. Vector competence is estimated experimentally by exposing mosquitoes to the virus when blood feeding. When the virus is found in the abdominal gut of the mosquito, it is considered to be infected. If the virus is found in the salivary glands and the saliva of the mosquito, then it is considered to be capable of transmitting the virus. Vector competence is shown to be dependent on both average temperature and diurnal temperature range, representing within-day temperature variance [18]. As a matter of fact, recent research has shown that the changes in diurnal temperature range are more important than the changes in average temperature for transmission of dengue (and other vector-borne diseases) [18,32,33]. A recent study examined the combined effect of the changes in temperature and diurnal temperature range on the epidemic potential of dengue worldwide over a 200-year period (1901–2099) using historical and predicted climate data for the high greenhouse gas emission scenario [34]. The study found a consistent increase in the vectorial capacity in many sub-tropical and temperate areas, amplified due to the changes in diurnal temperature variability, while it was decreased in the hottest areas of the globe, including the Saharan desert, the African Sahelian region, and the Middle East. The study estimated an optimal epidemic temperature at 29.3°C when the diurnal temperature range was zero. At lower and higher temperatures, the vectorial capacity was found to reach high levels when the diurnal temperature variation would allow parts of the days to be close to the optimal temperature. These findings indicate that short-term temperature fluctuations and extreme temperatures need to be considered when investigating the impact of longer-term changes in the climate on the transmission dynamics of mosquito-borne diseases. Lastly, it is important how long it takes for the vector to become infected to be able to transmit the virus, which is known as the extrinsic incubation period, yet another temperature dependent variable.

A growing body of research has investigated the associations between climatic factors and arboviral disease risk across space and time. Research findings indicate significant lag associations between temperature and rainfall and disease risk, as well as spatial variation in these associations [17,35–38]. Of particular importance has been research on the relationship of ENSO to arboviral diseases [3,39]. A recent study in Sri Lanka found a strong association between the Oceanic Niño Index (ONI, an ENSO index) and local weather patterns, and also ONI and local dengue risk at 6 months of lag in a highly endemic district [40]. A follow-up study showed that this increased risk was likely to be mediated by increased vector breeding [41]. The original study, furthermore, showed that increasing weekly mean temperature (above 29.8°C) and increasing cumulative rainfall (above 50 mm) were significantly associated with increased dengue risk at lags of 4 and

6 weeks, respectively. This type of information can facilitate an early and cost-effective response to abnormal disease events, including outbreaks, by providing public health authorities with sufficient time to plan and deploy vector control interventions, such as source reduction, environmental management, larvaciding, and adulticiding, to suppress vector proliferation [41]. Even if a safe vaccine becomes available for dengue or other arboviral diseases in the future, vector control is likely to remain a key complementary strategy for their control [42]. Shorter lag times can, however, be anticipated in warmer climates where the development rates of vectors and viruses are faster. The study concluded that the spatial variation observed in dengue risk could be due to the differences in human mobility and behavior, land use, and intensity of vector control interventions.

2.3. Eco-evolutionary effects of climate change and variability on disease transmission

In a more general ecological perspective on habitat suitability, climate change will affect ecological systems on a wide front [43–45]. This is expected to disturb the currently settled network of ecological interactions, which are well-known to drive species abundance and geographical distribution [46–49]. Food-web dynamics will likely undergo shifts that may have complex consequences on the abundance of *Aedes* mosquitos [50]. Resource competition between mosquito species is also expected to have equally significant effects on abundance. In broader terms, climate change will disturb ecological niche spaces in which current ecological systems have settled, and ecological communities will accordingly respond to novel niche-space geometries [45,51–53]. This will likely result in complex dynamics when climate change affects a multitude of species, with a cascade of changes in interspecific interactions.

More difficult to foresee at the longer time horizon, ecological reorganization can trigger concomitant evolutionary responses [51,52,54–56]—in both vectors [57] and pathogens [58,59]. While vectors are predicted to develop novel geographical distributions due to warmer climates [60], these distributions may not necessarily be perfectly linked to temperature, but can develop in a more complex way as a result of evolutionary adaptations to changing ecological systems to which they will be exposed. This suggests that pathogens and their primary vectors could be exposed to a novel set of mosquito species, which might imply shifts in resource-competition between mosquito species, but also potential for pathogen adaptation to new vector species. Such pathogen adaptation can be very fast due to high evolutionary rates [61], as in the case of increased chikungunya transmission rates in *Ae. albopictus* over only a few years [62]. Similar examples exist for other flaviviruses, such as the West Nile Virus [63]. With short generation times, extremely high mutation rates, potentially very large population sizes, and relatively simple bio-molecular machinery, RNA viruses like dengue and chikungunya are suited to display significant evolutionary change over relatively short-time scales [55,61,64]. More frequent and widespread disease outbreaks, as a result of increased vectorial capacity and basic

reproduction number (R_0) with global warming, may promote increased rates of evolution in pathogens owing to the increased size of and variation in exposed populations. Accordingly, virulence or transmissibility may change more rapidly; pathogen adaptations to additional vectors may become more likely; pathogen genetic variation may overall increase; and the risk of emergence of novel strains and serotypes may be expected to rise with climate change. While any of these consequences could involve serious epidemiological challenges, the actual and precise outcomes of the complex interplay between climate change, ecology and evolution in a changing socio-economic setting are not easily predicted [54,65] and need further scrutiny.

3. Research limitations and gaps

3.1. Timescale issues

Most of the work regarding future transmission dynamics of dengue and chikungunya is based on climate change projections for the end of the century. Although temperature projections are considerably less uncertain than for other variables such as rainfall, other timescales contribute to explain the total variability present in the variable of interest. Today's climate models still do not adequately represent these interactions (e.g. [66,67]), and hence a better understanding of cross-timescale interference and their impacts on the onset, magnitude and spread of dengue and chikungunya epidemics is urgently needed.

Furthermore, since predictive skill varies at multiple timescales [68], a formal skill assessment is needed to really evaluate how useful these forecasts are for decision making. This assessment depends on the timescale, the location, and the variable to be predicted (e.g., environmental suitability versus incidence).

Once operational forecasts systems dealing with multiple timescales are in place, further research is also needed to explore ways to put these different forecasts together, something that is known in the climate community as a seamless prediction approach (e.g. [20,69]).

3.2. Multi-driver influences

The influence of climate change and variability on infectious disease spread and emergence can, however, not be viewed in isolation as it is expected to interact with other drivers, such as urbanization, land use, and human mobility. Particularly, the unprecedented increase in human mobility, both at the local and global scales, is considered a major driver for the expansion of dengue and chikungunya [13,70]. A recent study in Indonesia demonstrated the role of human mobility in the intra-urban spread of dengue across 45 neighborhoods in Yogyakarta city [12]. Another study assessed the risk for chikungunya virus importation into France and Italy using air passenger journeys from areas with active chikungunya transmission and quantified the risk of onward transmission during the 2017 outbreak [8]. The derived risk maps combining vectorial capacity and human mobility had a good sensitivity in identifying at-risk areas for autochthonous transmission, with implications for targeting surveillance and outbreak response

activities. Therefore, a comprehensive analysis of the impacts of climate change and variability should consider the effects of non-climatic drivers to assess the changes in exposure and vulnerability to arboviral diseases in any given population.

3.3. Eco-evolutionary processes

A limitation in current state-of-the-art models of *Aedes* range expansions is the understanding of the effects of species interactions, which is often a primary determinant of the abundance and geographical distribution of species. Therefore, species interactions would be important to consider onwards. Evolutionary processes and their consequences with respect to both vectors and pathogens are currently poorly understood. One promising way for advancement would be through model development in the direction of including adaptive evolutionary principles to eco-epidemiological arboviral systems.

4. Expert opinion

This review highlighted the most critical research gaps that limit the predictive skill of arbovirus risk models and the development of early warning systems. It is not a trivial task to disentangle the role of different timescales and drivers. Nevertheless, previous studies have conducted timescale decompositions [71] to shed light on the particular weights of anthropogenic climate change versus natural climate variability in the case of the recent Zika epidemic in South America [72], and malaria in Africa [73]. Overall, the results suggest that the long-term climate change signal has a more important role for surface temperature (~40%-50% of the total explained variance) than for total rainfall (~0%-20% of the total explained variance), the particular values depending on the season of interest and the region of the world. Hence, considering only one timescale (e.g., the long-term climate change) is in general not enough, and the results might be misleading, especially when the dynamics of disease transmission – at least on the climate side – depends on specific thresholds of variables, which respond to different drivers and timescales. In other words, considering only climate change scenarios to try to project future behavior of dengue and chikungunya might work well only in those cases where the role of rainfall, humidity, and other key climate variables is negligible compared to surface temperature.

Future studies are also needed to ascertain the dependence of vectorial capacity parameters on temperature and diurnal temperature variations and further improve our currently modest understanding of these relationships. Short-term variability in temperature, as well as extreme temperatures, may affect the ability of mosquito vectors to effectively transmit pathogens. A number of studies have shown that diurnal temperature variations are more important than changes in average temperature for dengue transmission [18,34,74]. For instance, such understanding is particularly important for *Ae. albopictus*. This highly competent vector has expanded its geographic range drastically over the past decade, now also found in temperate areas in Europe, and has led to dengue and chikungunya outbreaks in previously disease-free areas [8]. In relationship to temperature, studies that incorporate

urban heat island effects are needed. The differences in temperature between cities and their surrounding areas can be substantial because of the generation and accumulation of heat as a result of urban construction and human activities. This phenomenon is known as the urban heat island effect. It is widely recognized that *Aedes aegypti*, the main vector of dengue and chikungunya, is well adapted to urban settings. Nevertheless, studies estimating future potential impacts from climate change have not included urban heat island effects in projections, despite the differences in temperature between rural and urban areas. In urban settings, urban heat island effects may lead to significant changes in the local climate, compared to projected global climate change in the twenty-first century. Particularly, under the influence of urban heat island effects, vectors and viruses can establish themselves and circulate locally in settings where the overall regional climate is not sufficiently conducive, for example Europe, the US and China, in the near future. Since these are precisely the locations where human population tends to aggregate, further research on this topic is urgently needed.

The interactions between climatic and non-climatic drivers of arboviral diseases are complex and dynamic. In the case of dengue, socio-economic factors such as poor housing quality, limited access to safe water and sanitation, and poor waste management are likely to exacerbate the effects of climate and weather in densely populated and highly connected urban settings. Longer-term changes in climatic conditions and seasons can further affect mosquito vectors, human activities, and land use. Such changes in natural environment, coupled with human mobility, can affect the distribution of mosquito vectors and pathogens, for instance by increasing the environmental suitability for vector proliferation, as well as population density. Socio-economic setting may limit the capacity of public health systems to respond to these changes, which can negatively affect the transmission risk of arboviral diseases. What is immediately apparent in this discussion is that most combinations of these drivers tend to increase vector abundance and host-vector interactions, compared to the potential effects of each driver individually, with only a few combinations reducing the risk of arboviral diseases [75]. The complexity implies that the effects of climate and weather on transmission risk will vary markedly by disease and by geographic location in the face of non-climatic drivers [76]. A comprehensive analysis of the impacts of climate change and variability should consider the effects of non-climatic factors to better understand the changes in exposure and vulnerability to arboviral diseases in a population. Nevertheless, because these factors contribute to overall effects individually, but also often interact and operate synergistically, predictions using individual factors should be interpreted cautiously.

Admittedly, studying arboviral diseases is complex. Transmission results from a complex web of dynamic interactions between humans, vectors, and viruses. These interactions are mediated by climatic and environmental factors operating at multiple geographic and temporal scales, and are also affected by human mobility and other non-climatic drivers. Furthermore, the eco-evolutionary processes acting simultaneously on vectors and pathogens in relation to hosts shape the present state of these interactions. Over the last few decades, it has become increasingly

apparent that effective prevention and control of vector-borne diseases should consider the effects of climate change on ecological systems. Understanding the changes in ecological systems in relation to climate change will require a better understanding of the potential changes not only in the physical environment but also the physiological, ecological, and evolutionary responses to these changes by foremost vectors and pathogens, and to some extent also hosts. We are currently limited in our understanding of eco-evolutionary processes and their consequences with respect to both vectors and pathogens, which may lead to unexpected increases or decreases in disease transmission. Future studies considering these different aspects are urgently needed to improve our understanding of complex disease dynamics in a rapidly changing climate and environment. An important outcome of such studies would be enhanced ability to promulgate early warning information, initiate adequate response, and enhance preparedness and response capacities on the ground.

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Declaration of interest

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